

英検 2 級

過去問

長文問題集

③

収録内容

2014年度第1回～2017年度第1回

3

A

次の英文 **A**, **B** を読み、その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

Turn It Off

The development of smartphones and other portable devices that connect to the Internet has certainly made our lives easier. With these devices, we can look at our e-mails, contact our friends, get information, prepare reports, and even do Internet banking from almost anywhere. But although they are useful, they can also cause problems. They make it difficult, for example, to (**26**) work-related e-mails. Moreover, some people become addicted to their devices, looking at them for new e-mails every few minutes. As a result, their stress levels rise, and they find it hard to concentrate on other things.

One solution is to create spaces where people cannot connect to the Internet. (**27**), more and more of these are being created around the world. In one big department store in London, for example, there is now an area called the Silence Room, where smartphones and other devices do not work. Some restaurants in the United States, too, have begun to ask their customers to leave their devices at the entrance when they go in. These restaurants believe this creates a better atmosphere for all customers. For many people, however, such breaks from the Internet are too (**28**). Experts say that, for these people, an even better idea is to take a vacation where they spend even more time away from the Internet.

Surveys show that most people remain connected to the Internet during ordinary vacations. The majority also check their work e-mail every day. This means that the vacation is not really a rest from their usual lives. Now, a number of travel companies and hotels have started to offer vacations where devices such as cell phones and computers cannot be used at all. On these vacations, traditional activities such as playing games, reading, and outdoor adventures are offered. Sometimes, experts are available to help people break their addiction to the Internet and to advise them on other ways to use their time. Such vacations allow people to experience some of the pleasures of a (**29**) life.

- | | | | | |
|------|--------------------------|---------------------|------------------------|------------------------|
| (26) | 1 send out | 2 search for | 3 escape from | 4 cover up |
| (27) | 1 Nevertheless | 2 Indeed | 3 Otherwise | 4 Unfortunately |
| (28) | 1 strict | 2 costly | 3 strange | 4 short |
| (29) | 1 less relaxing | | 2 more modern | |
| | 3 less convenient | | 4 more artistic | |

Plants and Music

As farmers and gardeners know, plant growth is affected by many different factors. Genes, light, temperature, and humidity can all influence the size of a plant. What about music, though? Some scientists, both amateur and professional, have been reporting for decades that music can help plants grow better. However, other scientists (**30**) these reports. After all, plants have neither ears nor brains, so how can they be affected by music? One explanation that has been suggested is that the chemicals in plants respond physically to the different sound waves that we hear as music.

Around 20 years ago, French scientist and musician Joel Sternheimer argued that he had found a method of influencing the chemicals in plants by playing certain musical notes. Sternheimer claimed he could use this method to control both the size and the (**31**) of tomatoes. Sternheimer said that when he played the notes to tomatoes, they grew two and a half times bigger than they normally did and that they became sweeter. Sternheimer also claimed to have helped tomatoes resist disease by using different notes, which changed the balance of the chemicals in the plants.

Recently, a team of South Korean scientists led by Mi-Jeong Jeong has done research that seems to support Sternheimer's claims. (**32**) them, certain genes in plants are affected by certain notes. The team carried out an experiment in which they played different pieces of classical music while looking at the activity of genes. They found that two genes became more active when certain notes were played. Based on this, Jeong believes that it may be possible in the future to (**33**) the growth of plants. Some scientists have questioned the results of the experiment. If these results are true, though, this would be good news. Using sounds rather than chemicals to help plants grow would be not only cheaper but also much less harmful to the environment.

- | | | | | |
|------|----------------|-------------|---------------|--------------|
| (30) | 1 doubt | 2 include | 3 need | 4 publish |
| (31) | 1 color | 2 taste | 3 shape | 4 number |
| (32) | 1 According to | 2 Thanks to | 3 In spite of | 4 Except for |
| (33) | 1 improve | 2 match | 3 record | 4 measure |

Brain Training

Much research has shown that the human brain is affected by how it is used. This has caused many people to believe that we may be able to make our brains work better by training them. Just as we go to the gym and do physical exercise to make our bodies stronger, perhaps we could do mental exercises to make our brains stronger. In fact, a number of companies now sell computer programs which, they say, allow us to do exactly this. But are such training programs worth spending money on?

Until recently, there has been a lack of proper scientific research into the effectiveness of these programs. In response to this, BBC Lab UK, a British website, decided to carry out a study of the effects of brain-training computer programs on overall brainpower. In the study, 13,000 people took part in a six-week program in which they did brain-training exercises three times a week. At the end, their mental skills were tested by scientists. The conclusion was that, while the participants got better at the particular exercises they did, their overall brainpower did not increase.

Other research, though, shows that these programs can have a beneficial effect on the brains of older people. A recent study carried out by a team at the University of California, Los Angeles, researched the effects of a six-month program on 69 people with an average age of 82. These people were asked to play a series of games involving memory, language, and mathematics. At the end of the six-month program, the participants showed clear signs that their brains were working better.

Associate Professor Michael Valenzuela of the University of Sydney says the evidence so far shows that people under the age of 60 are not likely to benefit from brain-training computer programs. Rather, for these people, the most effective method of keeping the brain working properly is to keep the body healthy. Moreover, although such programs may help older people, the evidence suggests that the most important thing for people over 60 is doing a lot of different activities. Valenzuela points out that activities such as meeting people or playing chess, which do not cost money, do as much for the brain as brain-training computer programs.

(37)

Recently, some companies are selling computer programs that they claim

- 1 give people a way to learn new skills by doing physical exercise.
- 2 train people to be able to perform more effectively at work.
- 3 strengthen people's brains by having them do mental exercises.
- 4 create images of people's brains that show how the brain works.

(38)

What was the purpose of the research done by BBC Lab UK?

- 1 To discover why the brainpower of certain participants did not increase.
- 2 To test scientifically how effective brain-training programs actually are.
- 3 To find out which age groups are better at different mental exercises.
- 4 To develop a new way of using computers to help people's thinking skills.

(39)

How was the study at the University of California, Los Angeles, different from BBC Lab UK's research?

- 1 It had many more participants than the BBC Lab UK study did.
- 2 It investigated the effects of brain-training programs on younger people.
- 3 It tested the memories of people who are good at math and languages.
- 4 It showed that brain-training programs help brainpower in some cases.

(40)

Associate Professor Michael Valenzuela says that

- 1 younger people should use brain training to avoid memory problems later in life.
- 2 people under 60 can best take care of their brains by staying physically healthy.
- 3 people should avoid doing too many activities at once because it can stress the brain.
- 4 older people are less likely to benefit from using computers than younger people are.

Freerice

In 2007, John Breen, a professional computer programmer living in the United States, had a clever idea. Breen, who is the father of two boys, wanted to create a website that would help teenagers study. At the same time, he felt very concerned about the situation of hungry people around the world. He wanted to find a way for people, especially teens, to help solve the problem of world hunger without having to spend their own money. One day, he came up with a way to achieve both of these goals at once.

Breen's idea was to create a website which he called Freerice. This website contains a simple game designed to help young people learn more English vocabulary. A word is shown on the screen with four other words below it. The player has to choose the one that has the closest meaning to the first word. Each time the player answers a problem correctly, the website donates 10 grains of rice to the United Nations World Food Programme (WFP), an international organization that helps people suffering from food shortages. The amount of rice that a player succeeds in donating is shown on the site.

The donations of rice are paid for by charging companies a fee for putting advertisements on the site. Each time a correct answer is chosen, an advertisement appears below the problem. The companies agree to pay a small fee each time the advertisement is shown, and this money is then used to buy rice. When the player chooses the wrong answer, the right answer is shown. Later, the same problem is shown again to check whether the player has learned the word. In this way, the game both teaches those who play and helps people who need food.

From the beginning, the website has been very successful. Within a year, players were donating up to 100 million grains of rice a day. In 2009, Breen gave the website to WFP, which added many new subjects, such as math, foreign languages, and literature, as well as versions of the game in other languages. Breen and others believe that the secret of the site's popularity is that players enjoy not only trying to get a high score but also donating as much rice as possible.

(41)

In 2007, John Breen wanted to

- 1 make more children in the United States interested in donating money.
- 2 let Americans know about the problem of hunger in poor countries.
- 3 teach English to children in poor countries around the world.
- 4 find a way for young people to both learn and help others.

(42)

What do we learn about the website called Freerice?

- 1 It has been teaching children in developing countries how to farm rice.
- 2 It gives food to people who answer the most questions.
- 3 It can be used to provide food to an international charity.
- 4 It was created by the United Nations World Food Programme.

(43)

How does Freerice earn money for its activities?

- 1 By allowing companies to sell their games on its website.
- 2 By asking players to buy rice in order to use the website.
- 3 By charging players a small fee every time they make a mistake.
- 4 By showing advertisements when players answer correctly.

(44)

The United Nations WFP

- 1 put games for more subjects onto the site and made it available in more languages.
- 2 gave as much rice as possible to countries that had teams competing on Freerice.
- 3 produces more than 100 million grains of rice a day and gives it to the poor.
- 4 has started giving an award to the player who visits the website the most.

(45)

Which of the following statements is true?

- 1 The games that Breen created for Freerice were not very popular with teenagers at first.
- 2 The idea for Freerice was first thought of by two boys who were living in the U.S.
- 3 Some people think Freerice is successful because its users can enjoy helping people.
- 4 Charities get donations even when visitors to Freerice's website do not play the games.

次の英文[A], [B]を読み, その文意にそって(26)から(33)までの()に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

A Better Wet Suit

Sharks are often thought of as dangerous fish that attack and kill people. In the past, though, they very rarely (**26**) human beings. In Australia, where there are many sharks, on average only about four people a year used to be attacked. Over the past several decades, however, the number of shark attacks has gone up dramatically. One reason for this change is that there are more people participating in water sports, and many people feel worried about their safety. Now, an Australian company called Shark Attack Mitigation Systems (SAMS) has a (**27**). They have designed wet suits which may help prevent attacks by sharks.

The new designs are the result of a two-year project carried out by the company and scientists from the University of Western Australia. Recent research has shown that, although sharks use their sense of smell to hunt, they mainly use their eyes when attacking. Sharks' eyes, however, are different from those of humans. (**28**), sharks cannot see colors. SAMS used this research, together with computer software, to figure out how different patterns would appear to sharks underwater. Using these results, they have created two different kinds of wet suit.

The first kind is designed mainly for surfers. This suit has thick black and white stripes that make the surfer stand out clearly. The design makes it look as though the surfer would be dangerous or unpleasant to eat. The other kind of suit, which is designed mainly for divers, works in almost the opposite way. It has a pattern on it that is similar to the way the sea looks to the shark. This means that the shark is less likely to (**29**) the diver. In other words, the wet suit should make divers invisible to sharks. Tests using sharks have found that, although they attack ordinary wet suits, they do not attack either of the new suits. SAMS's suits have already gone on sale, and the use of them is expected to reduce the number of shark attacks in the future.

- | | | | | |
|------|--------------------|-----------------------|-------------------------|-------------------|
| (26) | 1 bothered | 2 observed | 3 ignored | 4 guided |
| (27) | 1 question | 2 shortage | 3 responsibility | 4 solution |
| (28) | 1 Otherwise | 2 For instance | 3 Nevertheless | 4 If so |
| (29) | 1 trick | 2 escape | 3 notice | 4 avoid |

The Baby Box

Finland has one of the highest standards of living in the world, but in the early 1900s, it was a very poor country with many social problems. One of these was the high number of babies that died shortly after birth. In fact, every year around 6.5 percent died. Now, Finland has one of the lowest death rates for babies in the world. Many people believe that the main reason for this (**30**) is the country's policy of giving a free box of essential baby goods to every woman who is going to have a baby.

The box contains clothes, blankets, a mattress, toys, picture books, and other items that the government believes every baby needs. The boxes were first introduced in 1938 for poor mothers. Then, in 1949, they began to be given to all mothers. (**31**), mothers appreciated the boxes because they helped them to save money. However, Finnish people today are much wealthier and now like them for a different reason. They are very busy, and the boxes help them to save time preparing for the new baby.

The contents of the baby box have changed over time. Sometimes this reflected changes in (**32**). In the beginning, for example, the boxes contained cloth for making baby clothes. At that time, most mothers made baby clothes themselves. Later, this was changed to ready-made clothes. The boxes have also been used to (**33**) mothers' behavior. In the past, the boxes contained bottles for giving milk to babies, but recent research suggests that it is healthier for babies to drink their mothers' milk. To encourage mothers to stop using bottles, the government is no longer putting them in the boxes. Though the boxes have changed, for most Finnish mothers, receiving one remains an essential part of having a baby.

- | | | | | |
|------|--------------|-----------|---------------|-------------|
| (30) | 1 cost | 2 failure | 3 decrease | 4 criticism |
| (31) | 1 Originally | 2 Instead | 3 By contrast | 4 In short |
| (32) | 1 leadership | 2 weather | 3 diet | 4 lifestyle |
| (33) | 1 forgive | 2 defend | 3 punish | 4 influence |

Cradle to Cradle

Since the Industrial Revolution, the lives of many people around the world have been greatly improved. But although modern methods of manufacturing goods have helped many people to get out of poverty, they also damage the environment. As people have become more aware of environmental issues, they have begun to look for better ways of making things. What people are searching for is an approach to producing goods that will also help us to live in harmony with nature.

Two thinkers who have had a big effect on this search are William McDonough, a designer, and Michael Braungart, a chemist. In a book published in 2002, they suggested an approach which they called Cradle to Cradle (C2C). With C2C, the materials used in manufacturing are limited to two types—technical and biological “nutrients.” Technical nutrients are things such as metal and plastic, but only those that can be used again and again without harming the environment. Biological nutrients, such as paper and leather, come from plants or animals and can be returned to nature.

Based on their approach, McDonough and Braungart developed a system for rating goods according to how close they come to the C2C ideal. In the system, goods are measured in various ways, including how safe they are for the environment and the amount of energy used to make them. In fact, over 400 products have been given good ratings under this system, from bricks used in buildings to baby goods. Additionally, one sportswear company has produced a whole range of C2C shoes which are 100 percent recyclable.

Now McDonough and Braungart have come up with a new approach which uses “upcycling.” This refers to recycling goods into new and better products. In this approach, the newly created goods would also help make the environment better. One example is a carpet made in Holland which can both be recycled and help clean the air. McDonough’s architecture company has also designed a building which creates more energy than it uses. The authors believe upcycling will help correct some of the damage that we have already done to the environment.

(37)

The introduction of modern methods of manufacturing has

- 1 made the lives of many people all over the world better.
- 2 led to inventions that help people solve environmental issues.
- 3 helped some people to get rich and caused others to become poor.
- 4 caused more people to search for ways to live without buying goods.

(38)

What is one thing William McDonough and Michael Braungart say in their book?

- 1 Designers should limit the number of technical nutrients used in their products to two.
- 2 Publishers should try to plant new trees to replace the ones they cut down to make paper.
- 3 People should only buy products that are biological and can be returned to nature.
- 4 Metals and plastics should only be used if they can be recycled without damaging the environment.

(39)

What is one way in which the Cradle to Cradle system measures goods?

- 1 It tries to figure out how safe they are for use in buildings.
- 2 It considers how much energy is needed to create them.
- 3 It looks at the price of the materials from which they are made.
- 4 It compares them with similar products and chooses the best one.

(40)

In the new approach, “upcycling” should be used to

- 1 build houses and offices entirely from recycled materials.
- 2 create products that have a positive effect on the environment.
- 3 find items that do not cause any air pollution when they are made.
- 4 make sure that we only buy items that have been previously used.

Working for the Future

Most people in the United States, in theory, work on average eight hours a day, five days a week. In truth, though, many people work much longer than this. Indeed, according to one survey, 86 percent of men and 67 percent of women work more than this each week. And although the five-day workweek is standard, some experts say it makes more sense for people to only work four days and enjoy a three-day weekend each week.

According to supporters of the four-day workweek, reducing the number of days people work has many advantages. Because employees have three days in which they can rest, they tend to be healthier and more hardworking when they are at work. Moreover, they take fewer holidays from work to do things like visiting the doctor or the bank. Another advantage is that employers can close buildings for the day and save money on lighting and heating. This is also better for the environment. In addition, people do not need to travel to work as much, and this reduces car use.

The biggest experiment with a four-day workweek started in the state of Utah in 2008. The economy was doing poorly, and the state needed to cut its spending. Governor Jon Huntsman, Jr. surprised everyone when he announced that most Utah public employees would only work from Monday to Thursday. Each Friday, about 900 state buildings were closed and state vehicles were left in their garages. As a result, Utah saved millions of dollars. Employees preferred the new system, even though they had to work for 10 hours on the days they worked. Additionally, carbon emissions were cut by 14 percent.

However, in 2011, a law was made that brought back the old system. This was done because some Utah residents thought it was inconvenient that government offices were only open four days a week. Although the experiment came to an end, some local cities continued with the new system. According to experts, in the future there will be a strong need to reduce energy use to help fight global warming. Also, many economists say there is likely to be low economic growth. For these reasons, the four-day workweek seems likely to become more common in many countries, even in private companies.

(41)

What is true about many people in the United States?

- 1 They want to work over eight hours per day but are not allowed to.
- 2 They work much longer than what is thought to be average.
- 3 They are able to take at least three days off every week.
- 4 They are being asked to take more days off during the week.

(42)

What is one thing that happens when people have a four-day workweek?

- 1 They take fewer days off for things like medical appointments.
- 2 They have trouble getting back on schedule after the long weekend.
- 3 Companies cannot get their employees to work hard at the office.
- 4 Companies are able to sell buildings that are not being used.

(43)

In 2008, Governor Jon Huntsman, Jr.

- 1 announced that it would be illegal to drive downtown on Fridays.
- 2 created a law that required companies to cut carbon emissions.
- 3 shut down government buildings on Mondays and Thursdays.
- 4 introduced a new idea for saving money in the state of Utah.

(44)

Why did Utah decide to end its experiment in 2011?

- 1 Cities decided to stop using the state's system and create a new one.
- 2 The experiment caused economic growth in the state to slow down.
- 3 Some residents wanted government offices to be open five days a week.
- 4 People were more worried about global warming than working conditions.

(45)

Which of the following statements is true?

- 1 There are fewer public holidays in the U.S. than there are in other countries.
- 2 Governments are less likely than private companies to adopt shorter workweeks.
- 3 Working longer hours for fewer days has been shown to help the environment.
- 4 Women in the U.S. have been working 67 percent more hours than men have.

3

A

次の英文 **A**, **B** を読み, その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

The Science of Cooking

In 2006, a British magazine called *Restaurant* listed the top three chefs in the world: one in Spain, another in France, and the third in Britain. What all of these chefs had in common is that they had been influenced by a new approach to cooking called “molecular gastronomy.” This term was invented in the 1980s by Nicholas Kurti, a physicist, and Hervé This, a chemist, to describe the scientific study of the (**26**) of food.

Of course, scientists have been studying food for a long time. However, they have mainly studied how different foods affect human health, rather than how cooking methods affect the flavor and other qualities of food. Over time, the ingredients used in cooking have changed greatly. (**27**), the methods used to cook them have stayed largely the same. Hervé This has especially focused on studying common cooking beliefs, such as that salt should always be added to water when cooking green vegetables. According to him, these ideas often have no basis in scientific fact.

Hervé This has also been studying the effects of different flavors on one another and coming up with suggestions for new combinations of ingredients. It is this aspect of his work that has had the biggest (**28**). For example, Heston Blumenthal, a chef who has been particularly influenced by Hervé This, has become famous for his unusual dishes. Bacon and ice cream served on toast, or oysters and fruit jelly, may sound strange at first. Yet they have proved very popular with both critics and customers. Hervé This, however, believes that his research may someday have an even more (**29**) use. The reason is that his research could help solve the global problem of people eating unhealthily. He hopes that, by making healthier foods taste better, molecular gastronomy will encourage people to make food choices that can prevent serious diseases.

- | | | | | |
|------|----------------------|--------------------|----------------------------|-----------------------|
| (26) | 1 vitamins | 2 shortages | 3 cost | 4 preparation |
| (27) | 1 In addition | | 2 On the other hand | |
| | 3 Therefore | | 4 Fortunately | |
| (28) | 1 recovery | 2 chance | 3 impact | 4 loss |
| (29) | 1 important | 2 dangerous | 3 private | 4 entertaining |

Ancestors Online

In the United States, many people want to know more about their family histories. Finding out about one's ancestors, though, was often very difficult in the past. Nearly all Americans have ancestors who were immigrants from various countries. Because of this, their family histories are often (**30**). Moreover, because people moved around a lot, records for one family were often kept in churches, local government offices, and libraries in different parts of the country. However, new developments in technology have made learning about one's ancestors much easier. For this reason, (**31**) in this area has increased among ordinary people.

By far the biggest factor behind this has been the growth of the Internet. Many government records have now been put online and people can search these databases for the names of relatives. The biggest of these is the 1940 census, a study in which officials visited every house in the United States and asked detailed questions about the people who lived there. The (**32**) can now be accessed for free on the Internet. Many people use this information to find out more about their ancestors. There is also a good deal of information on the Statue of Liberty–Ellis Island Foundation website, which provides information about the millions of passengers who arrived in New York Harbor to settle in the United States between 1892 and 1954.

Another development has been social networking sites. Many people use these sites to contact relatives they have never met. These relatives often have more information on common ancestors, including photographs. They can also introduce people to other relatives. These sites make it possible for people to contact family members that they have never heard of before. (**33**) these developments, more and more people are filling in the details of their family histories. In fact, more than 9 million Americans now describe researching their ancestors online as their hobby.

- | | | | | |
|------|-----------------|---------------|---------------|--------------|
| (30) | 1 disappointing | 2 complicated | 3 helpful | 4 available |
| (31) | 1 trouble | 2 tourism | 3 interest | 4 employment |
| (32) | 1 results | 2 experiments | 3 designs | 4 courses |
| (33) | 1 Compared with | | 2 In spite of | |
| | 3 Apart from | | 4 Thanks to | |

Landfill Mining

Human beings have always produced waste, and they have often gotten rid of it by burying it in certain locations in the ground. This way of dealing with waste has continued in modern times, although the amount of waste being produced has increased enormously. However, putting waste into landfills, as these sites are now called, sometimes causes poisonous substances to get into the surrounding area. Landfills also take up valuable land that could be used for other purposes.

In recent years, though, people have been paying more attention to what is known as “landfill mining.” This means removing waste from landfills in order to recycle metals and other materials that have been buried in them. As resources such as metals become harder to find and more expensive to buy, a number of companies are finding the idea of landfill mining attractive. At the same time, governments want the poisonous chemicals contained in many landfills to be removed and treated. Nevertheless, landfill mining remains uncommon because it is still relatively expensive.

Now, though, a British company has come up with a new approach to landfill mining that could make it much more profitable. The company, Advanced Plasma Power (APP), is building a plant next to an old landfill in Belgium. This landfill contains 16 million tons of waste that have been buried there since the 1960s. The plant will remove any metals or other materials that can be recycled and then change the remaining waste into a new type of gas called “syngas.” This gas can then be burned as fuel to make electricity, which can then be sold.

APP believes that its project will make a lot of money as well as be good for the environment near the landfill. According to the company, about 45 percent of the buried waste can be recycled. The remainder, when it is turned into gas, will provide enough power for more than 60,000 homes. Moreover, the process will remove all poisonous chemicals from the waste. APP says that, as resources become more expensive and the price of electricity rises, this kind of project will become more and more common. It will also create space for new landfills or allow the land to be used in new ways.

(37)

Burying waste to get rid of it

- 1 is becoming less common because there is not enough room.
- 2 is an idea that was not thought of until modern times.
- 3 can cause poisons to be released into the local environment.
- 4 can improve the value of land in areas with too much waste.

(38)

Why are some companies attracted to the idea of removing waste from landfills?

- 1 It is the best way to get certain metals that are illegal to mine.
- 2 It is much cheaper than buying materials made of recycled products.
- 3 They are being offered money by governments to clean up the waste.
- 4 They can dig up materials that are becoming rare and expensive.

(39)

How will Advanced Plasma Power use some of the waste from landfills?

- 1 By turning it into a special gas which can then be used as fuel.
- 2 By selling it to people so they can burn it to heat their homes.
- 3 By adding metal to it to make a new kind of building material.
- 4 By creating islands with it which can become sites for power plants.

(40)

APP says that its approach will

- 1 reuse the metals and other materials to build new power plants.
- 2 result in almost half of the waste buried in the landfill being recycled.
- 3 create a new kind of landfill that is safer for the environment.
- 4 provide enough materials for thousands of new homes to be built.

Saving Guano

The use of fertilizers to help crops grow is necessary for agriculture. One of the most effective organic fertilizers in the world is known as “guano.” This means the droppings of birds and bats. These droppings are full of nitrogen and other chemicals that help plants grow. Huge amounts of guano left by seabirds used to be common in various parts of the world, but especially on islands off the coast of Peru in South America.

Although the value of guano as a fertilizer was known to local people in Peru, it was not until the 19th century that it began to be used in other countries. As people realized how effective guano was, a huge trade grew up. Hundreds of workers went to the islands and collected the guano, loading it into boats that then took it abroad to be sold. Soon, guano became the main export of Peru. However, after cheaper artificial fertilizers were developed by chemists in Europe in the 1900s, guano became less popular.

In recent years, though, guano and other organic fertilizers have started to grow in popularity again. One reason for this rise in demand is that many artificial fertilizers have risen in price, which has made guano more economical for farming. Another reason is the increasing demand for organic food in wealthy countries, because the chemicals that are used in man-made fertilizers can be harmful to human health. Moreover, such fertilizers often damage the environment. Guano, however, is completely organic.

Unfortunately, just as the guano industry has begun to improve, a new threat to it has appeared. The seabirds that produce guano mainly eat a small kind of fish called anchoveta that used to be much more common in the sea off Peru. Anchoveta, though, are also often used to feed animals that are being raised for meat. As the consumption of meat has increased around the world, the demand for anchoveta has also grown, and now fishing boats from many countries come to the area to catch them. Both the fish and the birds might disappear from the area if the fishing is not controlled, and Peru is working hard to get countries to agree to limit the number they catch.

(41)

The fertilizer known as guano

- 1 is created when seabird droppings combine with things in the soil.
- 2 is good for crops grown on farms but not for other kinds of plants.
- 3 used to be found only on the islands off the coast of South America.
- 4 contains a lot of chemicals which are good for plant growth.

(42)

Why did the use of guano go down in the 20th century?

- 1 It was difficult to find workers who were willing to dig it up.
- 2 It became harder to transport because fewer boats were being built.
- 3 People began to doubt how effective fertilizers were for growing food.
- 4 Scientists began to create new types of fertilizers that were less expensive.

(43)

What is one reason that guano has increased in popularity recently?

- 1 More people in wealthy countries want to buy organic food.
- 2 Companies can grow more vegetables when using natural fertilizers.
- 3 Food grown with guano has a better flavor than other food.
- 4 Artificial fertilizers have been made illegal in many places.

(44)

What is one new threat to the seabirds that produce guano?

- 1 The number of boats coming to Peru to hunt them has risen recently.
- 2 Fishermen are destroying the places where the seabirds live.
- 3 Many of the fish they eat are being caught for the meat industry.
- 4 Some of them are being killed by farmers to feed their animals.

(45)

Which of the following statements is true?

- 1 The price of guano is so high that local people in Peru can no longer buy it.
- 2 Peru must protect both anchoveta and the seabirds to save the guano industry.
- 3 The trade that developed around guano caused other industries in Peru to lose money.
- 4 Traditional guano collecting in Peru is more harmful to human health than first thought.

3

A

次の英文 **A**, **B** を読み, その文意にそって (26) から (33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Plastic from Air

One major cause of global warming is the release of carbon dioxide by companies and individuals into the air. Because of this, researchers around the world are trying to find new ways to reduce the amount being (26). Some ideas include using fossil fuels more efficiently or finding new sources of energy. Another solution has been to find ways to trap and store carbon dioxide. It is also possible, however, to make the carbon in carbon dioxide into materials.

One example of a company that has begun to make materials from carbon dioxide is Newlight Technologies, which was formed by two friends, Mark Herrema and Kenton Kimmel. They found a way to separate the carbon from carbon dioxide. They then used the carbon to create a kind of plastic. This plastic has various advantages. First of all, the process of making it removes carbon from the atmosphere. (27), unlike traditional plastics, it quickly breaks up and is absorbed into the environment after it is thrown away. Also, it is cheaper than regular plastics, which are made from oil. This could make it popular with manufacturers. In fact, the new plastic is already being used in various ways, and demand for it is increasing. Because of this, the owners are now (28) the business.

Other companies are using carbon dioxide to make other sorts of materials. One company creates a soft material which is made stronger by carbon dioxide from the air. It can be used in buildings, shoes, and other goods. Another company has found a way to create a kind of stone by combining carbon dioxide with other substances. This could be used in buildings or in roads. All of these materials have the advantage that they are cheap and, in turn, attractive to manufacturers. Because of this, they seem likely to become some of the most (29) ways of dealing with carbon dioxide.

- | | | | | |
|------|-----------------------|--------------------|-----------------------|----------------------|
| (26) | 1 destroyed | 2 purchased | 3 produced | 4 buried |
| (27) | 1 Nevertheless | 2 Therefore | 3 Occasionally | 4 Moreover |
| (28) | 1 explaining | 2 closing | 3 expanding | 4 selling |
| (29) | 1 successful | 2 costly | 3 difficult | 4 educational |

The Business of Giving

Nat Sloane is a British venture capitalist. This means that his job is lending money to new companies, and he often gives them advice on how to make their businesses grow so that they can make more money. However, Sloane is also interested in charities, and he donates money to some of them. One day in the 1990s, he realized that some of the advice he gave to companies would actually be (30) for charities. Together with a friend, he founded Impetus-PEF. This organization asks businesspeople to assist charities by giving them expert advice for free.

Charities are there to provide aid to people, but they often have little idea about ways to increase their income or how to spend the money they get more efficiently. This is where people with business experience can be helpful. In many countries today, governments are spending less than they used to on programs for people who need help. (31), charities and other voluntary organizations need to provide more aid than before. In fact, the advice that is given by Impetus-PEF leads to about 25 percent growth in the charities' income, and, even more importantly, to 30 percent growth in the number of people the charities can assist.

A good example is the charity called the St Giles Trust. This organization works to find homes and jobs for former prisoners. Before Impetus-PEF was (32), St Giles Trust was very small. It only worked with 2 prisons, and it had only found housing for 160 people and jobs for 8. After Impetus-PEF began giving it advice, though, it grew dramatically. It is now working with 24 prisons, and it has found housing for 2,407 people and jobs for 226. This is one example of how the Impetus-PEF approach to advising charities has proven to be (33). Because of its achievements, this approach is now being adopted throughout Europe.

- | | | | | |
|------|-------------------|------------|-----------------|--------------|
| (30) | 1 useful | 2 strange | 3 average | 4 impossible |
| (31) | 1 At most | | 2 As always | |
| | 3 For this reason | | 4 In particular | |
| (32) | 1 involved | 2 replaced | 3 damaged | 4 paid |
| (33) | 1 risky | 2 honest | 3 effective | 4 negative |

Left-Handed Competition

Most human beings are either right-handed or left-handed, although there are a few who can use both hands equally well. Nobody is sure why most people are one or the other, but experts agree that a mixture of genetic and environmental factors determines which we will be. Another mystery is why so few people are left-handed. Research shows that only about 10 percent of people are left-handed, and it has been this way for at least 5,000 years.

Many scientists believe that the ability of human beings to cooperate—for example by sharing tools—has played an important role in our evolution. From this point of view, it is natural that we have evolved to use the same hand because it allows people to share tools more easily. Why, then, are some people left-handed? One idea is that, when people fight against one another, being left-handed can be an advantage because the opponent does not always know what to expect. Recent research by Daniel Abrams and Mark Panaggio, two mathematicians at Northwestern University in the United States, has given support to this theory.

The two researchers used mathematics to predict how many people in a certain sport would be left-handed. In some sports, such as boxing, participants only compete against one other opponent. In others, such as American football, cooperation between the members of a team is also extremely important to winning. The researchers predicted that there would be many more left-handed athletes in the first kind of sport than in the second. When they examined actual sports, they found that their predictions were accurate. For example, about 25 percent of boxers are left-handed. By contrast, the percentage of left-handed American football players is much lower than the percentage in the rest of the population.

The researchers believe their results can be applied to human evolution as a whole. It seems that, in human evolution, it was very important that the majority of people be right-handed. This is because most of the time cooperation was more important to our survival than competition. However, in certain situations, competition was more important. In these cases, having about 10 percent of the population left-handed was an advantage for human beings.

(37)

Experts agree that

- 1 most children are born with the ability to use both hands equally well.
- 2 there are several factors that cause people to be either right- or left-handed.
- 3 the percentage of left-handed people has been increasing for 5,000 years.
- 4 some children begin using one hand more than the other at about 10 years old.

(38)

What do scientists think is one reason that most humans have evolved to be right-handed?

- 1 Using the same hand improves people's ability to compete against others.
- 2 Using the same hand helps people to understand mathematics better.
- 3 Using the same hand allows people to share tools with each other easily.
- 4 Using the same hand makes it easier for people to become leaders.

(39)

What is one thing that Daniel Abrams and Mark Panaggio discovered?

- 1 Sports without teamwork have a larger percentage of left-handed participants.
- 2 American football requires cooperation between 25 percent of the players.
- 3 There was a little less cooperation in boxing than they had first expected to find.
- 4 The theory that they had about left-handed people was very unreliable.

(40)

What does the research on left- and right-handed people suggest?

- 1 It is becoming more and more important to have left-handed people in society.
- 2 People who are born left-handed can train themselves to become right-handed.
- 3 There is little evidence that people became left-handed as a result of evolution.
- 4 Having some left-handed people in the population has helped humans to survive.

Industrial Strength

Like other developed countries, the United States has gone through a number of economic stages in its history. Originally, it was an almost entirely agricultural economy, but by around the time of the Civil War in the 1860s, manufacturing had become more important than agriculture. Then, by the late 20th century, the service industry had reached the same level as manufacturing. Since then, the manufacturing industry has declined dramatically. Indeed, so few people work in factories now that many economists say the U.S. is a “post-industrial” economy.

There were two reasons for this trend. The first was the development of new technology, especially computers. Because of this, far fewer workers were needed to make the same number of goods. The other reason was rising wages in the U.S. These made American workers much more expensive than workers in developing countries such as China. As a result, many American companies moved their factories abroad to take advantage of the cheaper labor. On the other hand, within the U.S., service industries, such as tourism and healthcare, grew greatly.

Now, though, there are signs that some of the manufacturing industry that went abroad is beginning to return. One example of this is the clothing industry. Originally, many companies in the clothing industry moved their factories to countries in Asia and South America to take advantage of the lower wages. Since then, however, wages have risen in these countries. Additionally, the rising price of fuel has made transporting the clothes back to the U.S. much more expensive. Furthermore, many American customers say that they would rather buy clothes made in the U.S. to support the economy, even if they were to be more expensive.

However, when companies reopen their factories in the U.S., they are faced with a big problem. Because most of the clothing jobs went abroad years ago, there are few American workers with the skills and experience needed to operate the machines. As a result, many companies have begun to set up training programs to teach workers the skills they need. If these programs work, it seems that the U.S. may begin to see many types of manufacturing increase again.

(41) What happened to the United States around the time of the Civil War?

- 1 The number of jobs available in factories stopped growing.
- 2 Its economy went from being agriculture- to manufacturing-based.
- 3 It began to be called “post-industrial” by economic experts.
- 4 The service industry became the most important source of jobs.

(42) Rising wages in the U.S.

- 1 allowed workers to afford new technology such as computers.
- 2 caused some shops to close because they could not pay their workers.
- 3 caused companies to move factories to places where workers were paid less.
- 4 allowed the development of technology that could take the place of workers.

(43) What is one reason that clothing manufacturing is returning to the U.S.?

- 1 The price of fuel needed to import clothes made abroad has gone up.
- 2 People feel that the quality of clothing is better if it is more expensive.
- 3 The clothing made in Asia does not match American tastes in fashion.
- 4 Workers in factories in South America are being treated badly.

(44) Why are companies that are trying to reopen factories in the U.S. having trouble?

- 1 There are not enough workers who know how to use the machines.
- 2 It is too expensive to run training programs for new factory workers.
- 3 The weakness of the economy has caused the U.S. to raise taxes.
- 4 The equipment used in manufacturing abroad must be brought back.

(45) Which of the following statements is true?

- 1 Experts expect that factory jobs in Asia will continue to decline dramatically.
- 2 Not only clothing but other types of industries might return to the U.S. in the future.
- 3 No human workers will be needed at all in the future to make goods in factories.
- 4 Wages in developing countries might soon become higher than those in the U.S.

3

A

次の英文 **A**, **B** を読み、その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び、その番号を解答用紙の所定欄にマークしなさい。

City Farming

Urban farming is becoming increasingly popular across the United States. Urban farmers grow plants and raise animals for food just like farmers in the countryside. (**26**), there are some differences. For instance, they do this on empty pieces of land in towns and cities. Urban farms can be of any size—from large ones created on the sites of former factories to small ones in people's backyards. It is estimated that around 20 percent of urban households produce some of their food by farming.

One reason for this trend is that people are concerned about how farming in the countryside has changed. They believe that modern farming methods, which use many chemicals, are damaging nature. They also think that it is wrong to use fossil fuels to carry food to cities because doing so is one of the causes of global warming. Many people also think that the food produced using modern farming methods is (**27**). They want to grow vegetables and raise animals themselves, because they can be sure that the food they eat does not contain any chemicals that may harm their bodies.

People also choose urban farming for (**28**) reasons. Many families can grow food in their own yards. Thus, they do not have to buy fruits and vegetables at the store. Moreover, people without their own yards can rent small pieces of land in towns and cities very cheaply. Although urban farming has many benefits, some people are concerned that keeping animals in crowded cities may lead to people catching diseases from them. Supporters of urban farming, though, say that diseases among animals are (**29**) on farms in the countryside. This is because in the countryside thousands of animals are kept together in small spaces, so it is easy for diseases to pass from one animal to another. Considering these advantages, many families are turning towards urban farming.

- | | | | | |
|------|---------------------------|----------------------|------------------------|-----------------------|
| (26) | 1 Traditionally | 2 Furthermore | 3 Unfortunately | 4 Nevertheless |
| (27) | 1 dangerous | 2 attractive | 3 unenjoyable | 4 steady |
| (28) | 1 moral | 2 economic | 3 historic | 4 official |
| (29) | 1 more comfortable | | 2 more common | |
| | 3 dirtier | | 4 safer | |

Rewriting DNA

Recently, some people have started a new hobby that is called biohacking. Biohacking is when people who are not professional scientists alter the DNA of various living things. They do this for fun or to learn new things about DNA and genetics. Before, only organizations like universities and companies could purchase the equipment and materials to do this. Now, however, cheap materials and equipment are (30) on the Internet. As a result, people can create laboratories in their homes and experiment on DNA.

Biohackers are sometimes even able to solve problems that professional scientists have not been able to solve. A good example is Daniel Burd. For a while, scientists had been trying to find a way to reduce the amount of waste caused by plastic bags, which normally would remain in landfills for hundreds of years. (31), the 16-year-old schoolboy Daniel did what many scientists could not. Using only simple household items, he grew bacteria which can eat plastic bags. As a result, Daniel was awarded a prize and a university scholarship for his project.

Most biohackers carry out experiments at home. They buy DNA or other biological materials over the Internet and experiment with them. One biohacker created bacteria that turns green if it finds a poisonous chemical in yogurt. A team of biohackers is creating bacteria that can live in the mouth and protect our teeth. In this way, biohackers are trying to create useful new organisms that can help people. Some people, though, are (32) this trend. They believe that amateurs may create harmful new diseases. But defenders of biohacking argue that amateurs like Daniel Burd can come up with ideas that professional scientists are unlikely to think of. They believe that, in the future, many important (33) will come out of the biohacking movement.

- | | | | | |
|------|-----------------|--------------|-----------------|----------------|
| (30) | 1 disappointing | 2 available | 3 rare | 4 valuable |
| (31) | 1 As usual | 2 At first | 3 Amazingly | 4 Additionally |
| (32) | 1 tired of | | 2 encouraged by | |
| | 3 pleased with | | 4 worried about | |
| (33) | 1 documents | 2 complaints | 3 commercials | 4 discoveries |

Green Taxis

Like many large cities, London faces a number of difficult environmental problems. One of these is the greenhouse gases being released into the air by cars. Another is the huge amount of used cooking oil being put into the city's sewer system* by restaurants and other food businesses. Now, though, Uptown Oil and Pure Fuels, two small companies in London, are trying to solve both of these problems at the same time. These companies are collecting used cooking oil and turning it into a clean fuel that can be used to run cars.

The two companies collect used cooking oil from restaurants, bars, and cafés, and then turn it into a clear, eco-friendly biofuel. When used in cars, this fuel releases 80 percent less carbon dioxide than ordinary gasoline. It also produces about 60 percent less of the gases that can harm people's health. The companies' main customers are London's taxis. These taxis visit the companies' fueling stations in order to fill up on the biofuel.

The two biofuel companies together regularly serve about 1,000 taxis. However, in London, there are 21,000 taxis altogether, as well as millions of ordinary cars. At the same time, large amounts of cooking oil are still released into the sewer system every year. Indeed, the local water company spends about \$1.5 million every month cleaning the city's sewer system so that it does not become blocked with the oil, which would cause flooding and other problems.

Despite the benefits of biofuels, few companies have started producing them. According to Uptown Oil, there are two major problems that make it difficult for companies to recycle cooking oil into biofuel. One of these is the government's policies. The government has very strict rules controlling companies that want to provide fuel for cars. These rules make it very expensive to establish a new company. The second major problem is that, at present, only a few cars have engines that can run entirely on biofuel. If these difficulties can be removed, however, making biofuel from cooking oil could turn into a big business.

*sewer system: 下水道

(37)

Two small companies in London are trying to

- 1 make both the air and the local sewer system cleaner.
- 2 help restaurants that have no way to get rid of their kitchen waste.
- 3 invent cars with special engines that run on cleaner fuel than gasoline.
- 4 save more money for both the transportation and food businesses.

(38)

What is one advantage of the clear biofuel?

- 1 It can help taxis run much more quickly than gasoline can.
- 2 It does much less harm to the environment than regular fuel.
- 3 People can buy it at restaurants instead of at gas stations.
- 4 People use it to produce eco-friendly foods for cafés.

(39)

The local water company in London

- 1 is looking for ways to pay for damage caused by flooding.
- 2 is paying drivers to use fuel made from waste.
- 3 spends a lot of money every month cleaning up oil.
- 4 says that too much car oil has been getting into the water.

(40)

What is one challenge faced by companies that want to make biofuel from recycled cooking oil?

- 1 Government rules make it hard to start new companies that provide fuel for cars.
- 2 It is difficult to compete with big businesses that already have many customers.
- 3 They cannot afford the type of engines that are able to run on biofuel.
- 4 The number of cars that can use biofuel will probably not increase in the future.

Growing Up in Australia

Since 2004, Australia has been holding its largest study of children's health ever. This project, "Growing Up in Australia," is tracking two groups of children from early childhood until they become adults. The project is being funded by the Australian government and involves hundreds of researchers visiting the homes of 10,000 children. Researchers hope to find a relationship between the children's future behavior and their health, education, and family environments.

When the study began, the first group of children was aged 0–1, while the second group was aged 4–5. In order to make sure that children from every kind of environment and background were chosen, the study found volunteers from each region. Some of these locations are in the countryside, where there are relatively few people, while others are in big cities. The volunteers for both groups were chosen from a variety of different social and ethnic backgrounds to represent the diversity of Australia.

The researchers visit the participants' homes every two years to gather data. They interview the parents about parenting styles and the child's daily activities. They also take measurements of the children's health and how well the children perform in school. The researchers also take note of social factors, such as the income of the child's family, the school the child attends, and the kind of community in which the child lives. They plan to use all of this information to improve children's lives.

Although the study has not been completed yet, researchers have already begun to publish some interesting results. For instance, after examining data from the second group of children when they were 10–11 years old, researchers found that children who spend a lot of time doing organized activities, such as music lessons or sports, have good control of their emotions and behavior. The study also found that they are able to complete their homework without their parents around. On the other hand, it found that children who watch a lot of television get upset easily and have worse behavior. The Australian government is now using this information to make better educational and health policies for Australian children.

(41)

What is one thing we learn about the study “Growing Up in Australia”?

- 1 Children will study hard to get educational funds from the government.
- 2 The government will pay health workers to improve children’s health.
- 3 Researchers have been gathering various data about children over a long period.
- 4 Specialists have been employing children to collect data about the environment.

(42)

What is true about children participating in the study?

- 1 They will represent their families in an international competition.
- 2 They will learn social skills from regional volunteers.
- 3 They come from many different family backgrounds.
- 4 They have to go to various cities in Australia.

(43)

Researchers who visit the children’s homes

- 1 gather data about children in order to help them lead better lives.
- 2 go to the children’s schools to learn more about Australian education.
- 3 measure how well parents can solve their children’s problems.
- 4 provide health services to children living in the community.

(44)

As a result of the study, researchers found that

- 1 watching television has no effect on children’s emotions.
- 2 participating in organized activities has a positive effect on children.
- 3 children who watch a lot of television are bad at sports.
- 4 children who do organized activities like to control their friends.

(45)

Which of the following statements is true?

- 1 Australians want the government to spend more money on education.
- 2 The government is unhappy with the results of the study.
- 3 Older children have better control of their emotions than younger children.
- 4 The study follows two separate groups of children.

3

A

次の英文 **A**, **B** を読み, その文意にそって(26) から(33) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Love Locks

“Love locks” are little locks on which lovers write their names. For over 20 years, couples around the world have been attaching such locks to the sides of bridges. After that, the couple often throws the key into the river below the bridge. This is meant to be a symbol of the couple’s love, which they hope will last forever. No one knows for sure where this idea started. However, there are many (**26**). Some people think that it started in Italy, and others think that it started in China. Wherever it started, though, it is now a common practice in many countries.

Although the people who attach the locks think the idea is (**27**), they have been causing problems in a number of cities. Paris, in particular, has been struggling with the difficulties caused by these locks. Paris is a favorite holiday destination for lovers, so many of these visitors began attaching locks to bridges around the city. One especially popular bridge was the Pont des Arts. Before long, the locks on the bridge were as heavy as 20 elephants. The (**28**) caused some parts of the bridge to break off and fall into the river.

In the beginning, the city council did not want to act against the custom because the council knew how much visitors liked it. In 2014, though, two Americans living in Paris started a campaign to have the locks removed. Many local people agreed, saying the locks were ugly and could further damage the bridge. In 2015, the city cut all the locks from the bridge. Many visitors have (**29**) this decision. They believe that this action destroyed an important symbol of love in the city.

- | | | | | |
|------|---------------------|--------------------|-------------------|------------------|
| (26) | 1 challenges | 2 chances | 3 theories | 4 reasons |
| (27) | 1 reliable | 2 romantic | 3 costly | 4 strange |
| (28) | 1 weight | 2 length | 3 storm | 4 fire |
| (29) | 1 delayed | 2 respected | 3 reached | 4 opposed |

The Foldscope

Every year millions of people, especially in low-income countries, die from diseases because their doctors cannot identify what is making them sick. These diseases are caused by germs and other creatures that are too (**30**) to see. Because of this, doctors must use a microscope to examine the patient's blood and identify the cause of illness. If a doctor can quickly identify the disease and give the correct treatment, many lives can be saved. However, microscopes are very expensive and so there are not enough of them in poor countries. This means that patients often have to wait a long time before they can be given the right (**31**) by their doctors.

Recently, however, a scientist in the United States, Manu Prakash, came up with a possible solution to this problem. This is an invention called the Foldscope. The Foldscope is a microscope made out of folded paper with a small light, lens, and battery inserted into it. By using the Foldscope, it is possible to make something appear about 2,000 times its actual size. Doctors can use it to examine a patient's blood sample. In this way, they can (**32**) what is wrong.

The Foldscope has many advantages over normal microscopes. It is very easy to use and people can attach a smartphone to take photographs of the sample. Moreover, it is very tough and will not break even if it is dropped. (**33**), it is very cheap. One costs only about 50 cents. The Foldscope is now being tested in three countries. Experts hope that, if found effective, this invention will save millions of lives in poor countries.

- | | | | | |
|------|-------------------|------------|--------------|-------------|
| (30) | 1 fast | 2 small | 3 far | 4 bright |
| (31) | 1 equipment | 2 medicine | 3 insurance | 4 skills |
| (32) | 1 give up | 2 make up | 3 find out | 4 throw out |
| (33) | 1 Additionally | | 2 At first | |
| | 3 Disappointingly | | 4 Since then | |

The Whooping Crane

Many birds migrate. This means that every year they fly somewhere warm in the winter and then return to the place they came from in the spring. One mystery is how birds know how to travel such long distances without getting lost. Scientists have wondered for years whether they are born with the knowledge of where to go or whether they learn it after they are born. Now, researchers have found evidence that shows that one type of bird, at least, learns how to migrate.

The bird is the whooping crane, a tall, white bird that lives in the United States. In the 1940s, the number of these birds decreased to fewer than 25 individuals, and many people feared that they would die out. Thanks to efforts to save them, they have now increased to about 600. However, the birds are not very good at taking care of their young. Because of this, eggs are brought to a site in Maryland. There, scientists from the Whooping Crane Eastern Partnership raise the baby birds.

Later, the young birds are taken from Maryland to Wisconsin, where the scientists teach them how to fly. From there, they fly about 2,200 kilometers south to Florida to spend the winter. The first time, scientists lead the way in a small airplane. From the second year, though, the birds travel by themselves. For 14 years, the scientists have kept a careful record of every flight the birds have taken between Wisconsin and Florida. They have also recorded which birds fly together, the relationships between the birds, and the gender of each bird.

When scientists analyzed the data, they were surprised by the results. These showed that each bird took a very different route. However, as they got older, they flew more and more directly between Wisconsin and Florida. This continued until they were five years old. This indicates that the birds gradually learned the best route to take. Moreover, when an older bird was in a group, the whole group flew more directly. This suggests that the older birds taught the younger ones a quicker route. Scientists say that this evidence marks a big step toward solving the mystery of bird migration.

(37)

What is one thing that scientists want to know about birds?

- 1 How they are able to migrate successfully.
- 2 Why they migrate to warmer places.
- 3 When they start migrating each year.
- 4 Whether some birds fly faster than others.

(38)

What is one reason scientists from the Whooping Crane Eastern Partnership raise whooping cranes?

- 1 They want to observe how the mothers take care of their young.
- 2 They are worried that the birds will fall down many times.
- 3 Whooping cranes' eggs need to be watched constantly.
- 4 Whooping cranes are bad at raising their own young.

(39)

How do scientists from the Whooping Crane Eastern Partnership support young whooping cranes?

- 1 They help the birds learn how to survive in Wisconsin's winter.
- 2 They fly together with a group of birds that are the same gender.
- 3 They show the birds how to fly to Florida in their first winter.
- 4 They provide food for the cranes on their journey south.

(40)

What is one thing that the data suggested?

- 1 Most birds like to visit new places when migrating.
- 2 Most birds stop flying to Florida when they are five years old.
- 3 Younger birds prefer not to fly together with older birds.
- 4 Younger birds are able to gain an important skill from older ones.

Educating Finns

Finland, a country in northern Europe, has been ranked as having the highest quality of life in the world. In particular, Finland has been attracting attention for the success of its schools. Since 2000, the country often appears near the top of an international survey of educational standards called PISA. This survey measures the academic level of 15-year-olds around the world in science, math, and reading once every three years.

What surprises many people about Finland's success is that the country is so different from the others that come at the top of the survey. Those countries' schools generally emphasize competition between students and between schools, and they often have long hours of study, much homework, and frequent tests. The success of this educational style has led to more countries introducing such competition into their own education systems. Finland, however, has taken a different approach to education.

The Finnish government requires no standard textbooks or tests. Rather, teachers are allowed to plan lessons using whatever materials they think will be best for their students. Moreover, students are not required to do much homework. In fact, the average 15-year-old studies less than 30 minutes outside of school each day. A student's academic performance does not affect which classes or schools that student attends, and the country has almost no private schools. All these policies are the opposite of those being adopted by other countries. The aim of this system is to promote equality between students instead of focusing on high scores.

In order to create equal opportunities for each student, the Finnish government relies greatly on individual teachers and trusts them to make the best decisions for their classes. These teachers are, in turn, rewarded with high salaries and social status. This creates strong competition for teaching jobs, so only the most highly trained and qualified candidates can become teachers. This focus on students' needs and high-quality teaching has led to Finland's success. Seeing the strong results of these methods, some countries have begun to take a second look at their approach to teaching. Although education systems vary from one country to the next, all governments share the desire to provide the best education they can to their students.

-
- (41) What is one reason Finland has been attracting attention recently?
- 1 Finnish students regularly perform well on surveys of academic ability.
 - 2 The government has begun providing free education to all students.
 - 3 A group of students have been working together to improve their schools.
 - 4 Most of its 15-year-olds want to study in the fields of math and science.
-
- (42) Most schools in the countries at the top of the PISA survey
- 1 encourage their students to participate in sports to create competition.
 - 2 focus on competition and require students to study for many hours.
 - 3 believe that all students should receive the same education.
 - 4 doubt that Finland's education system is good for students.
-
- (43) What is one thing that is unusual about the Finnish education system?
- 1 Schools encourage children to get high scores on English tests.
 - 2 The Finnish government requires students to take many tests.
 - 3 Teachers are encouraged not to teach children from other countries.
 - 4 Students are not separated according to their ability or results.
-
- (44) What is one thing that has led to Finland's educational success?
- 1 It gives job training to students who need extra help.
 - 2 It rewards students who achieve high scores on tests.
 - 3 It requires all teachers to teach the same lessons.
 - 4 It values and trusts the abilities of its teachers.
-
- (45) Which of the following statements is true?
- 1 Most Finnish students receive high salaries after graduating.
 - 2 Finnish teenagers travel around the world to study.
 - 3 Many governments are copying Finland's system.
 - 4 Students in Finland have very little homework.

2

A

次の英文 **A**, **B** を読み, その文意にそって(21) から(26) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Music to My Ears

Music plays an important role in people's lives around the world. Singing, as it does not require any special tools or instruments, is a particularly popular way for people to enjoy music. Although many people sing for fun, few people know that singing actually (**21**). From helping people fight diseases to reducing stress, singing can make people's lives better.

Singing can also be seen as a type of exercise. When people sing, their bodies use a number of important muscles. It also helps people improve their breathing and strengthens both their lungs and hearts in the same way as aerobic exercise such as jogging. (**22**), some experts say that singing can actually be a good alternative to exercise for elderly people who need to improve their weak lungs and hearts. Furthermore, singing increases the chemicals in the body that make people feel good, which can help reduce anxiety and depression.

New research has shown that singing (**23**) can also have positive effects. Don Stewart, a researcher at Griffith University in Australia, surveyed elderly members of a chorus. He found that although they had the same health problems as other elderly people, they were actually happier than those who did not sing or those who sang alone. In addition, the chorus members visited the doctor less and took fewer medicines. After noticing these advantages, some governments are considering promoting choruses as a way to reduce the cost of medical care for the elderly.

- | | | |
|------|---|---|
| (21) | 1 improves memory
3 has many health benefits | 2 leads to heart disease
4 causes a number of problems |
| (22) | 1 In fact
3 Unfortunately | 2 At last
4 Traditionally |
| (23) | 1 once a day
3 at school | 2 in a group
4 with a microphone |

Washing Away

Most people have heard about the large number of plastic bottles and bags that are floating around in the world's oceans, harming fish and other animals. Mark Browne, an environmental researcher, examined the types of plastic that had washed up on various beaches around the world. He was surprised to find large amounts of plastic, such as polyester and acrylic, which are not used to make plastic bottles or bags. From this, he knew that this plastic waste must be (24).

Since polyester and acrylic are found in many types of clothing, Browne thought that small plastic fibers were probably being washed off people's clothes and flowing through the sewer system* to the ocean. Wanting to solve this problem, Browne talked to a number of clothing companies. He recommended that they make clothes that were stronger and less likely to lose plastic fibers. However, many of these companies (25) was the problem. Instead, they said that these plastics could be coming from many other things, such as blankets, rugs, or even paint.

Unwilling to give up, Browne found an engineer named Blair Jollimore. Jollimore had noticed that plastic fibers from his washing machine had been causing a problem with his water pipes at home. He was able to solve this problem by inventing a device that stops the plastic from leaving his washing machine. Browne encouraged Jollimore to sell this invention to large companies so that they could reduce the amount of plastic waste getting into the sewer system. (26), Browne hopes that the amount of plastic waste in the ocean will eventually decrease.

*sewer system: 下水道

- | | | |
|------|------------------------------|------------------------------|
| (24) | 1 difficult to remove | 2 coming from somewhere else |
| | 3 more dangerous than before | 4 spreading to new places |
| (25) | 1 misunderstood that acrylic | 2 accepted that polyester |
| | 3 suggested that plastic | 4 denied that clothing |
| (26) | 1 In this way | 2 Rather |
| | 3 Even so | 4 At first |

Snake Hearts

Many large snakes eat as rarely as once every three months. When they do eat, however, they eat whole animals, such as deer or wild pigs. In order to digest these large meals, the heart and other organs inside the snake's body suddenly grow much larger. The heart, for example, does this so that it can pump blood around the body much faster. This allows the snake to digest food more quickly. After the snake has digested the meal, the organs go back to their usual size.

One snake that does this is the Burmese python. American biologist Leslie Leinwand and her team at the University of Colorado Boulder decided to research exactly how the Burmese python manages to expand its heart so dramatically. The team found that certain chemicals in the snake's blood cause the heart to grow. When the team added the snake's blood to cells taken from mice's hearts, they were surprised to find that the cells also began to grow.

After further experiments, Leinwand removed the chemicals from the snake's blood and gave them to living mice. As a result, the mice's hearts grew dramatically. Leinwand believes that her research may be able to help humans with heart diseases. Human hearts grow larger in two ways, one unhealthy and one healthy. When people have high blood pressure, for example, the muscles in their hearts grow to help the heart push blood around the body. This kind of growth actually damages the heart and makes it weaker. On the other hand, when people do exercise, their hearts grow larger in a healthy way.

According to Leinwand's research, the chemicals in the snake's blood cause the heart to grow in the same way that exercise usually does. She believes that if she could make a drug from these chemicals, it could cure certain heart diseases in humans. She says that more research is needed and that it will be years before a drug could be developed. However, she believes that the discovery is very exciting because it shows that the chemicals taken from one creature can affect the heart of a completely different creature.

(30)

Many large snakes change the size of their hearts in order to

- 1 make more space for the whole animals that they eat.
- 2 better digest the blood of the animals that they eat.
- 3 be able to catch more food more quickly.
- 4 help them digest large meals more quickly.

(31)

What is one thing that Leslie Leinwand's team learned about Burmese python blood?

- 1 It has the effect of making mice's heart cells grow larger.
- 2 It can be used to increase the size of farm animals.
- 3 It expands dramatically when added to the blood of other animals.
- 4 It reduces the size of mice's hearts while the mice are sleeping.

(32)

What happens to the heart of a person with high blood pressure?

- 1 It becomes able to produce more blood to pump around the body.
- 2 It becomes weak and damaged as a result of becoming bigger.
- 3 It has to be supported by various muscles in the body.
- 4 It has to be given a special medicine to help it grow.

(33)

Leinwand believes that

- 1 humans should drink snake blood in order to live a healthy life.
- 2 the drug she makes will benefit snakes and other wild animals.
- 3 her research can be used to develop medicine for some heart diseases.
- 4 Burmese pythons can protect humans from diseases spread by mice.

The Handwriting Debate

The development of computers has had a huge effect on society. As a result, many schools around the world have introduced computer classes, and some encourage students to use computers in other classes, too. This not only allows students to take notes and complete assignments more quickly but also helps them gain the skills they need for jobs they will do in the future. However, the new focus on computer skills means that many schools in developed countries are spending much less time teaching students how to write by hand.

Surveys show that many parents are concerned about the fact that their children spend so little time developing good handwriting skills. In the past, learning good handwriting was considered a necessary part of a good education, and these parents feel children are losing an important skill. However, some education experts say such parents are too focused on the past. The world has changed, they say, and typing is now more important than writing by hand.

On the other hand, recent research has shown that writing by hand may help people learn more quickly. Karin James, a psychologist at Indiana University in the United States, studied the brains of children when they were writing letters by hand and when they were typing. She found that children who wrote letters by hand used an additional three parts of the brain that children who were typing did not. Virginia Berninger, a psychologist at the University of Washington, also found that the brain was more active when writing by hand than when using a keyboard. Even more importantly, she found that when children wrote by hand, they used a wider range of vocabulary and expressed more complex ideas.

Handwriting is also important because it increases the hand's strength and flexibility. In particular, it strengthens the small muscles in the hand and allows people to make very accurate, small movements. This can be very important in certain jobs, such as for doctors who perform operations. All this research suggests that schools should think again before focusing entirely on having children use computer keyboards to write.

(34)

Why do some schools encourage students to use computers?

- 1 It takes students less time to learn how to type than it does to learn how to write.
- 2 It allows teachers to spend less time working with students in their classes.
- 3 It helps students perform tasks in class faster and prepares them for future jobs.
- 4 It helps students learn by connecting them with other students around the world.

(35)

Many parents are worried about their children because they

- 1 believe that their children are not learning an important skill.
- 2 think that the world is changing and their children need computer skills.
- 3 are not able to teach their children how to use computers at home.
- 4 feel that spending so much time studying will harm their children's eyes.

(36)

Karin James found that

- 1 a combination of writing and typing is the best way to help children learn.
- 2 the brain has a hard time handling information when a person is writing.
- 3 children who are typing use fewer parts of the brain than children who are writing by hand.
- 4 most people prefer to write letters to their children by computer rather than by hand.

(37)

How can learning handwriting help doctors?

- 1 It encourages them to focus more on their patients when performing operations.
- 2 It gives them stronger muscles which help them carry heavy medical equipment.
- 3 It helps them do their job better by allowing them to see small movements more easily.
- 4 It improves their ability to move their hands accurately while performing surgeries.

(38)

Which of the following statements is true?

- 1 Typing helps students come up with more ideas and organize them better.
- 2 Children use a larger variety of words when writing by hand.
- 3 Good handwriting has recently become an important skill.
- 4 Teachers spend a lot of time preparing for classes that teach typing.

2

A

次の英文 **A**, **B** を読み, その文意にそって(21) から(26) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Local Foods

Over the past 20 years, more and more individuals have been buying locally produced food because it is better for both people's health and the environment. Seeing these benefits, some retirement homes in the United States have switched from packaged food made in large factories to homemade dishes made with local ingredients. They were surprised to find that this change (**21**) as well. They found that by preparing meals at the retirement home, they were able to reduce the overall amount of money spent on food.

Two of the major costs of food production are packaging and transportation. When people buy from local farmers, the farmers can spend much less money on both. (**22**), the farmers lower the price of their products. As a result, the retirement homes can save money while the local farmers are able to make a larger profit, allowing more money to remain in the area. Thus, the local economy also improves.

FutureCare Cherrywood is one retirement home which has started using locally produced food. Sandy Stallings, the food service director, says that one of the most important results is how residents now feel about their food. They are more satisfied with their food and happier in general. Of course the improved taste of the food is important, but Stallings believes that getting residents (**23**) has also led to this change. Residents help to decide which foods will be served and even grow some vegetables themselves. By buying locally, a number of problems can be solved at once.

- | | | |
|------|---------------------------------------|---|
| (21) | 1 had economic advantages | 2 caused medical problems |
| | 3 reduced environmental damage | 4 decreased the availability of food |
| (22) | 1 Like before | 2 In turn |
| | 3 At least | 4 For once |
| (23) | 1 involved in choosing | 2 interested in exercise |
| | 3 to eat vegetables | 4 to use fewer services |

Smelling Fear

The sense of smell is an important tool for animals that live together in groups. Dogs and wolves, for example, use smell to communicate with their families and with other groups. In a similar way, smell is important for humans. In fact, it also helps us (**24**). Some scientists say that without smell, we would have trouble recognizing family members and learning how to get along in a group. This is because smell is closely related to our memories.

Katherina Hauner, a researcher at Northwestern University in Chicago, did an experiment to learn about smell, memories, and feelings—in particular, fear. She showed participants a picture of a man while causing them pain. (**25**), she had them smell roses. Later, when the people saw the picture again or when they smelled roses, they became afraid. This is because they remembered the smell, the picture, and the pain together as one memory.

Before beginning her experiment, Hauner had believed that there was a way to use smells to help people get rid of their fears. She knew that when people see something that they are afraid of many times, they learn not to be afraid of it. In her experiment, Hauner had the participants sleep in a room that smelled like roses. While the participants were sleeping, the smell caused them to remember the picture in their dreams many times. When they woke up, they were less afraid of the picture. Hauner was (**26**). She hopes doctors will use these results to develop treatments for fears in the future.

- | | | |
|------|---|--|
| (24) | 1 improve academic ability
3 reduce stress easily | 2 develop social skills
4 forget pain quickly |
| (25) | 1 With luck
3 At the same time | 2 For this reason
4 By then |
| (26) | 1 not able to help them
3 not surprised by the results | 2 worried they might get hurt
4 ready to start new research |

The Threat of Tourism

As air travel gets cheaper, more and more people are visiting famous sites around the world. Although this increase in tourism brings economic benefits to the areas around these sites, tourists also cause unexpected problems. In particular, some famous works of art are being affected. This is because people's breath increases carbon dioxide and humidity levels. Gradually, these levels damage old paintings and other works of art.

One famous site facing this problem is the Sistine Chapel in the Vatican in Rome. The 500-year-old paintings, especially the famous ceiling by Michelangelo, are so popular that as many as 2,000 people may be viewing them at a time. In 1994, after noticing that the visitors' breath was damaging the paintings, the Vatican purchased an expensive air-conditioning system to protect them. However, the crowds continued to increase, so in 2014, the Vatican decided to limit the number of visitors to about 6 million a year.

Another site that faces a similar problem is the Mogao Caves in Dunhuang, China. These caves are full of beautiful Buddhist paintings and sculptures that attract thousands of visitors every year. Many of the artworks are very old and, as with the Sistine Chapel, the carbon dioxide in the breath of visitors is gradually damaging them. Originally, 40 of the 400 caves were open to visitors, but this number was reduced by half in 2014. In addition, the number of visitors allowed into the caves has been greatly reduced.

A different solution is being tried in the Ajanta Caves in Maharashtra, India. The caves also have many ancient Buddhist paintings in them, and these too are being damaged. In order to protect the paintings, visitors are quickly rushed through the caves. However, many visitors complained about the short time, saying they could not look at the paintings properly, so the local government built a visitors' center with exact copies of the caves. Visitors are allowed to study these copies for as long as they like. The local government hopes this will provide a good balance between protecting the paintings and giving tourists a good experience.

(30)

As the number of tourists increases,

- 1 unexpected economic problems occur among people living around famous sites.
- 2 the carbon dioxide and humidity in their breath harm the things they go to see.
- 3 air pollution caused by the carbon dioxide from airplanes increases.
- 4 people have trouble breathing because of the high levels of humidity.

(31)

In 1994, the Vatican

- 1 allowed only 2,000 tourists to look at its paintings by Michelangelo.
- 2 invited 6 million visitors to see its 500-year-old wall paintings on one day.
- 3 installed an air-conditioning system in order to make visitors more comfortable.
- 4 tried to reduce damage to its paintings by buying an air-conditioning system.

(32)

What is one thing that has been done to protect the Buddhist artworks in Dunhuang?

- 1 More of the Mogao Caves have been closed to visitors.
- 2 Visitors are being asked to avoid breathing too close to the paintings.
- 3 Some of the visitors are being taught new ways to preserve paintings.
- 4 The number of visitors has been reduced from 400 to 40 a day.

(33)

Why were some visitors to the Ajanta Caves unhappy?

- 1 The majority of the paintings have turned out to be copies.
- 2 There were not as many Buddhist paintings as they had expected to see.
- 3 They did not have enough time to look at the paintings inside the caves.
- 4 The long lines at the visitors' center have prevented them from seeing the paintings.

Purifying Plants

During the 1970s, as oil prices rose rapidly, the United States, like many other countries, had trouble producing energy cheaply. As the price of fuel rose, people looked for ways to reduce the amount of energy they used on heating and cooling buildings. One way to do this was to make buildings that let very little air in or out. In winter, these buildings keep the warm air from heaters inside for longer, and in summer, they do the same with the cool air from air conditioners. This greatly reduces the amount of money people spend on fuel.

Unfortunately, though, these new buildings had an unexpected side effect. Many modern materials, such as plastic, release small amounts of poisonous chemicals into the air. When air remains inside the buildings, the amount of these poisons gradually increases, and people who work or live in them begin to feel ill. This problem is known as sick building syndrome. One solution to this problem, however, came from a surprising area—space research.

NASA, the U.S. space agency, employed a scientist named Bill Wolverton to study ways to keep the air clean inside spaceships. Spaceships, of course, cannot get air from outside because there is no air in space. In order to study ways to keep the air in spaceships free from poisons released by modern materials, Wolverton created the BioHome, a completely closed building made entirely of artificial materials. When people entered the building, they would suffer from sore eyes and find it difficult to breathe. To solve this problem, Wolverton tried introducing various kinds of plants into the BioHome. He found that most of them were effective at helping to clean the air.

According to Wolverton's research, a plant can absorb chemicals through its leaves and then use them to feed itself. He wrote a number of books explaining which plants are most effective against the various chemicals found in the air inside buildings. He also developed a new kind of container for plants that makes them even more effective in removing poisons. Thanks to Wolverton's research, many buildings can save energy and keep their air clean by using plants.

(34)

Why did people start making buildings differently in the 1970s?

- 1 They were trying to reduce the amount of energy that was being used.
- 2 They were trying to make buildings that could have air conditioners in them.
- 3 They wanted to increase the amount of air getting in and out of the buildings.
- 4 They wanted to make homes warmer and more comfortable in the winter.

(35)

Sick building syndrome is caused by

- 1 modern materials on which bacteria and other unhealthy things can grow.
- 2 materials releasing poisonous chemicals that get trapped in buildings.
- 3 a reduction in the amount of air available for people living in the building.
- 4 an increased number of people living and working together in the same space.

(36)

Why did Bill Wolverton create the BioHome?

- 1 To provide a place for NASA to keep the plants that it will send to space.
- 2 To discover what types of poisons are created by artificial materials.
- 3 To learn more about the types of plants that can be grown in space.
- 4 To find ways to make the air in spaceships safe and clean.

(37)

What was one result of Wolverton's research?

- 1 Wolverton reduced pollution by teaching people to use energy efficiently.
- 2 Wolverton developed plants that are more effective at removing poisons from the air.
- 3 Wolverton wrote a book that recommends the safest building materials to use.
- 4 Wolverton created a container that helps plants reduce dangerous chemicals in buildings.

(38)

Which of the following statements is true?

- 1 Many buildings made of artificial materials have been rebuilt to prevent illness.
- 2 Plants are able to clean the air with their leaves while getting food for themselves.
- 3 The United States made a lot of money as a result of rising oil prices in the 1970s.
- 4 People doing space research commonly suffer from sore eyes and breathing problems.

2

A

次の英文 **A**, **B** を読み, その文意にそって(21) から(26) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Thinking of Stress

Stress is said to be one of the leading causes of death in modern society. When people get stressed, their hearts work too hard, leading to heart attacks and other diseases. However, Kelly McGonigal, a psychologist in the United States, is trying to change the way people commonly see stress. She says research shows that people with a lot of stress are more likely to die young but only if they think stress is bad. In fact, according to research at the University of Wisconsin, this (**21**) stress is actually killing about 20,000 Americans every year.

In one experiment at Harvard University, researchers observed how participants' hearts reacted to stress. At first, the participants' hearts beat faster and became tighter. However, once the participants were told that stress was a natural and good response to a difficult situation, their hearts began to react differently. Their hearts beat faster without becoming tight. The heart becoming tighter leads to heart disease and possibly death. (**22**), a rapidly beating heart causes no damage.

McGonigal also says that oxytocin, a chemical produced by the body during stress, may have a positive impact on the heart. While oxytocin is most known for its ability to make people happier and more social, it also encourages the heart to relax during times of stress. In fact, studies have found that it can even help the heart recover from damage. This means that stress can (**23**). McGonigal hopes people will learn that the way they think about stress can improve their experience of it.

- | | | |
|------|---------------------------------------|---|
| (21) | 1 rare idea about | 2 modern cure for |
| | 3 negative view of | 4 common cause of |
| (22) | 1 Unfortunately | 2 As a result |
| | 3 In addition | 4 On the other hand |
| (23) | 1 be good for people's health | 2 be a problem for relationships |
| | 3 improve the quality of sleep | 4 increase the heart's speed |

High-Altitude Wind Power

Wind power is a clean and renewable source of energy. However, few areas have the frequent strong winds needed to produce wind power. Wind near the ground is not a reliable source of energy because it is sometimes strong and sometimes weak. Unlike the wind near the earth's surface, the wind high in the sky is (24) nearly everywhere. Because of this, some companies are now building wind power plants at high altitudes, as they can obtain stable wind power almost anywhere.

Altaeros Energies is a U.S.-based company working to develop "high-altitude wind power" technology. They have designed a huge balloon that catches the wind's power at up to 600 meters in the air. The balloon is connected to cables which keep it in place and transport the energy to the earth's surface. As the wind is much stronger at 600 meters, these balloons can create more than twice as much electricity as traditional wind power at a lower cost. In other words, these balloons (25) to produce energy.

Not only do Altaeros's balloons reduce costs, they also solve a number of other problems associated with traditional wind power. To start with, the initial setup is quite easy. While traditional wind power plants require months of construction, the balloons' setup only takes a few hours. (26), they are so high in the sky that they do not disturb people's daily lives. For instance, they cause little noise pollution, and people do not have to see them every day. Seeing all these benefits, a number of large companies have invested in Altaeros's wind power balloons.

- | | | |
|------|---|---|
| (24) | 1 constant and powerful
3 weak but reliable | 2 clean and clear
4 strong but unstable |
| (25) | 1 reduce the wind needed
3 are a less attractive way | 2 increase the space needed
4 are a more efficient way |
| (26) | 1 Like before
3 Therefore | 2 At most
4 Furthermore |

SuperAgers

People around the world are living longer. One problem that affects many older people, though, is memory loss. Usually, this is just an inconvenience, but it can be a sign of brain disease. Scientists have been studying the brains of people suffering from memory loss in the hope of finding a way to treat the problem. Researchers at Northwestern University in Chicago, however, decided to take a different approach. They began looking at the brains of people who still have strong and healthy memories even as they get older.

The team of researchers calls these people “SuperAgers.” The researchers asked people over the age of 80 who believed they had unusually good memories to take part in the project. To determine if these people truly had good memories, the researchers first tested a group of people between the ages of 50 and 65. Then, they gave the same test to the people over 80 and compared the results. All those who scored as well as the younger group were considered to be SuperAgers. This, however, only amounted to about 10 percent of those who had volunteered.

After this, the brains of the younger group and those of the SuperAgers were scanned. The researchers were surprised to find that the brains of the SuperAgers looked just as healthy as—and in some ways healthier than—those of the younger group. In particular, the outer layer of the brain, known as the cortex, was thicker than the people in the younger group. This part of the brain is connected to both memory and thinking. Moreover, the part of the brain that helps people pay attention was also thicker than is normally the case with older people.

The SuperAgers not only had good memories but were also more energetic and active than other people the same age. This goes along with other studies which have shown that people who participate in many activities and are involved in their communities are less likely to suffer from brain problems. What scientists do not yet know, though, is whether such people are born with genetically stronger brains or whether it is their diet and lifestyle that keep their brains healthier.

(30)

What are researchers at Northwestern University doing about memory loss?

- 1 They are studying the brains of people who have good memories in their old age.
- 2 They are focusing on cases where memory loss is not caused by brain disease.
- 3 They are trying to reduce the effects of brain diseases on people's memories.
- 4 They are looking for a way to reduce the inconvenience of memory loss.

(31)

In the study, "SuperAgers" were defined as

- 1 people over 80 who answered at least 10 percent of the test questions correctly.
- 2 people over 80 whose memories are as good as people in their 50s and early 60s.
- 3 people between 50 and 65 whose memories are better than average for their age.
- 4 people between 50 and 65 who are likely to live a healthy life through their 80s.

(32)

What were researchers surprised to learn about the SuperAgers?

- 1 They got along better with younger people than with people of their own age.
- 2 They had better memories than others but are not as good at paying attention.
- 3 There were no differences between their brains and those of other people their age.
- 4 The part of their brains related to memory and thinking was unusually healthy.

(33)

What does evidence from a variety of studies suggest?

- 1 Some people are unlikely to be able to improve their memories through practice and exercise.
- 2 Most people with strong brains are genetically different from people with normal brains.
- 3 Living an active and social life is related to the health of people's brains.
- 4 Eating healthily before exercising has a positive effect on people's memories.

Sweet Treat

Chocolate, which is made from cocoa beans, is one of the world's best-loved foods. Some chocolate manufacturers, however, are warning that there may be a major shortage of chocolate in the near future. This is because the world is now consuming more chocolate than it can produce. Moreover, demand keeps increasing. In fact, the price of chocolate increased by over 60 percent between 2012 and 2014.

One reason for the chocolate shortage can be seen as a good thing. Chocolate is a luxury item, so most of it is consumed in wealthy countries. However, with the strong economic growth of many developing countries over the past two decades, demand for chocolate has risen. For example, the development of a large middle class in China and India has led to more chocolate being consumed in these countries. Demand for chocolate has also grown dramatically in South America. In this sense, the growing shortage of chocolate can be seen as a sign that the standard of living is rising around the world.

Another reason, though, is more troubling. Today, about 70 percent of the world's cocoa beans are produced in two West African countries, Ghana and the Ivory Coast. Production in these countries, though, has been declining. This is because there have been increasing periods of very dry weather caused by climate change. Moreover, cocoa trees are being threatened by various diseases that attack them. Because cocoa trees take a long time to grow and produce beans, these diseased trees cannot be replaced easily.

Cocoa bean farmers have been responding by developing new kinds of cocoa trees. These new kinds are stronger against disease and produce more beans than normal trees. However, progress is slow because of the time it takes for each tree to grow. At the same time, demand for chocolate is rising, so cocoa is becoming more and more valuable. Therefore, chocolate manufacturers are investing more money in the industry so that production will increase in the future. By doing this, they hope to avoid cocoa shortages. With luck, people around the world will continue to have access to one of their favorite foods.

(34)

What are chocolate manufacturers concerned about?

- 1 People prefer living a healthy life to eating sweets such as chocolate.
- 2 People will not want to buy chocolate any longer because it is too expensive.
- 3 They will have to increase the price of chocolate by 60 percent to cover costs.
- 4 They are not able to produce as much chocolate as people want to buy.

(35)

The change in demand for chocolate shows that

- 1 the quality of life of people around the world is improving.
- 2 many people living in South America are now selling chocolate.
- 3 middle-class people are unlikely to buy large amounts of chocolate.
- 4 Chinese people are more interested in luxury goods than other people.

(36)

In the area where the majority of cocoa beans come from,

- 1 climate change is helping to produce healthier cocoa trees.
- 2 it is taking longer than usual to grow new cocoa trees.
- 3 fewer beans are being produced because the weather is becoming drier.
- 4 beans are 70 percent smaller than beans grown in South America.

(37)

In order to keep up with demand for chocolate,

- 1 manufacturers are looking for ways to make sweets that taste like chocolate.
- 2 manufacturers are spending more money on increasing production.
- 3 farmers are moving to new places to grow cheaper cocoa beans.
- 4 farmers are buying cocoa trees that can produce beans more quickly.

(38)

Which of the following statements is true?

- 1 The number of diseases that threaten cocoa trees is decreasing.
- 2 Cocoa trees that can protect themselves against diseases are being developed.
- 3 People in China and India have stopped believing that chocolate is a luxury food.
- 4 Chocolate will probably become the world's best-loved food in the near future.

次の英文 **A**, **B** を読み, その文意にそって(21) から(26) までの () に入れるのに最も適切なものを **1, 2, 3, 4** の中から一つ選び, その番号を解答用紙の所定欄にマークしなさい。

Battling Bacteria

When people catch diseases that are caused by bacteria, they are often given drugs called antibiotics, which work to kill bacteria in the body. However, there are now many new types of bacteria that cannot be killed with antibiotics. One type of bacteria, called MRSA, has become (**21**) hospitals and nursing homes. MRSA spreads easily where there are many sick people in one place, and it cannot be treated using traditional antibiotic drugs. As a result, many scientists are looking for new antibiotics that can fight such bacteria.

One group of researchers at United Arab Emirates University has been looking to frogs for an answer. Frogs live in wet environments that have a lot of bacteria, so their skin has evolved to protect them from dangerous bacteria. Scientists have known about the bacteria-killing chemicals on frogs' skin for many years. However, there are some challenges to finding a way to use those chemicals. One is that the human body destroys the chemicals before they can act as a medicine. (**22**), some of the chemicals on the frogs' skin can harm human cells.

Michael Conlon, leader of the study, says that his team has discovered some solutions. They have made small changes to the chemicals on the frogs' skin to make them less harmful to humans, while also making them better at killing bacteria. Despite these advances, Conlon says that it will (**23**) drugs from these chemicals. Although these drugs may not be ready soon, it is expected that one day they will play a large role in fighting bacteria.

- | | | |
|------|---|--|
| (21) | 1 a new hope for
3 a big concern for | 2 less common in
4 easier to cure in |
| (22) | 1 Fortunately 2 Therefore | 3 In exchange 4 What is more |
| (23) | 1 be impossible to create
3 build demand for | 2 take time to develop
4 reduce the damage of |

Tipping in the United States

A tip is a small amount of money that customers give to someone who has provided a service for them. This payment is made in addition to the charge for the service. Tips are common in many countries, including the United States—where tips are given to waiters, taxi drivers, and other workers. In fact, in most American restaurants, waiters are paid a very low hourly wage. This is because it is expected that they will (**24**). As a result, tips are a very important part of their income.

Tipping has both good and bad points. Many restaurant owners like the system because they can pay waiters less and so charge lower prices. This is attractive to customers, who also believe that tipping encourages waiters to give better service. (**25**), tipping can sometimes be confusing for visitors to the United States. Even many Americans are unsure of exactly how much they should give in some situations, such as when a hotel worker carries suitcases to a guest's room.

Now, some people in the United States are trying to find ways to improve the situation. Jay Porter, a restaurant owner in San Diego, found that his waiters were arguing with each other about their tips. This was creating problems for his business, so he decided to (**26**) in his restaurant. Instead, he added a fixed service charge to each bill and raised the waiters' wages. After this, his workers were happier. Although anti-tipping policies are still rare in the United States, some people would rather avoid the problems that tipping can cause.

- | | |
|--|--|
| <p>(24)</p> <p>1 make money through tips</p> <p>3 return money to the restaurant</p> | <p>2 find a second job</p> <p>4 pay fewer taxes</p> |
| <p>(25)</p> <p>1 Furthermore</p> <p>3 On the other hand</p> | <p>2 Similarly</p> <p>4 To make matters worse</p> |
| <p>(26)</p> <p>1 hire new employees</p> <p>3 ask for advice</p> | <p>2 invite more customers</p> <p>4 get rid of tipping</p> |

Ballet Brains

Often after people spin around in circles, they begin to feel dizzy. The world seems to be moving around them, and it is difficult to stand properly. Usually, a liquid in their ears causes signals to be sent to their brains to help them keep their balance. However, if they spin and then stop, the liquid keeps moving, and they become confused about whether they are still moving or not. Ballet dancers, though, often spin around as they dance, but they do not suffer from dizziness.

A group of brain scientists at the Imperial College London decided to examine the brains of 29 female ballet dancers. They put each of the dancers in a chair that quickly turned round and round in a dark room. When they stopped the chair, they asked the dancers to record how long they continued to feel as though they were spinning. The scientists then did the same thing with a group of female athletes. They found that the ballet dancers felt dizzy for a much shorter time than the athletes did.

After they had finished the experiment, the scientists scanned the brains of all the women. To their surprise, they found that the part of the brain called the cerebellum was much smaller in the ballet dancers than in the athletes. The cerebellum is the part that processes the signals from the liquid in the ears. This suggested it is possible to avoid dizziness by changing the way the brain processes these signals. In fact, the ballet dancers' brains had actually developed over the years to make them stronger against dizziness.

According to the scientists, their research suggests the possibility of finding a new treatment for people who suffer from illnesses that cause them to become dizzy. Usually these diseases happen when something goes wrong with their ears. Until now, medical researchers have found it difficult to treat such problems. The new research suggests that, rather than trying to cure the problems in their ears, it may be better to focus on finding a way to help their brains adapt. In this way, they could become as strong as ballet dancers against dizziness.

(30)

People become dizzy after they spin around because

- 1 signals cause the brain to think that the body is moving even though it is not.
- 2 an object makes people lose their balance by blocking signals between the body and the brain.
- 3 they have been standing for such a long time that their body becomes tired.
- 4 the liquids that they have drunk cause their brains to feel like they are spinning.

(31)

What is one thing that brain scientists at the Imperial College London found?

- 1 Female ballet dancers were able to continue dancing as soon as they had stood up.
- 2 Female ballet dancers had trouble spinning around when they were in the dark.
- 3 Female athletes took less time to stand up from a chair than female dancers.
- 4 Female athletes continued to feel dizzy for a longer period of time than female ballet dancers.

(32)

The experiment suggested that ballet dancers' brains

- 1 are smaller and more easily damaged than athletes' brains.
- 2 can be cured of dizziness by scientists researching the brain.
- 3 produce more liquid to help them process signals more efficiently.
- 4 have developed a different way to process signals sent from their ears.

(33)

According to scientists, how can the research on ballet dancers' brains be used?

- 1 To reduce the amount of practice needed to become a good dancer.
- 2 To treat health problems that have been difficult to treat.
- 3 To help people who have trouble with their hearing.
- 4 To show people how to learn to concentrate better.

Cairo's Garbage

A major problem that faces modern cities is how to deal with the increasing amount of garbage that people produce. Cairo, the capital of Egypt, has a population of 15 million people. This leads to an enormous amount of garbage. Yet, until recently, the city had no official system for collecting garbage. Instead, a group of people known as the Zabbaleen collected and recycled the garbage in order to make a living for themselves. Now, many people are saying that the Zabbaleen's methods were more efficient than those used in most of the world.

The Zabbaleen are mainly the descendants of poor farmers who first settled in Cairo during the 1950s. Traditionally, the men went from house to house collecting garbage with a cart pulled by a donkey. Meanwhile, the Zabbaleen women stayed at home and sorted through the garbage that the men brought back. Any food that was collected was fed to their pigs and chickens, while other materials were used to make cloth and other goods. In this way, over 80 percent of the garbage was recycled.

Life was not easy for the Zabbaleen. Their housing was bad, and they often suffered from poor health. Over time, however, both their housing conditions and their lifestyles have improved. A school, clinic, and park were built by the government for the Zabbaleen, and the children started to receive a better education. More and more of the younger generation are able to receive university degrees. At the same time, they began to focus on recycling high-tech materials, such as computer parts, which has allowed them to make more money.

From the 1980s, Cairo began to produce more garbage than the Zabbaleen could collect. Then, in 2003, the city decided to employ two European companies to help handle some of the garbage. Many people were not pleased with the new system. One problem was that it cost more money. Another was that far less was recycled, leaving more garbage to be buried in the ground. Hoping to reduce costs and improve the environment, many people would now like to expand the traditional method of collecting garbage.

(34)

Until recently, the city of Cairo had

- 1 hired a group of people to collect and recycle garbage.
- 2 asked its population to make efforts to reduce the amount of garbage.
- 3 not provided an official service for dealing with garbage.
- 4 not had enough money to pay the people who removed garbage.

(35)

What did the Zabbaleen do with most of the garbage they collected?

- 1 They used it as food for their animals or turned it into new products.
- 2 They took it to garbage collection facilities using their donkeys and carts.
- 3 The women sorted it and the men took it back to where it came from.
- 4 The men used it to produce the materials they needed to build their farms.

(36)

What problem did the Zabbaleen face?

- 1 They had poor living conditions and often became sick.
- 2 They were unable to make enough money to feed their families.
- 3 The government did not allow them to get an education.
- 4 The government did not provide them with computers.

(37)

Why do many people want to expand the old style of garbage collection?

- 1 They want to find a way to support other developing countries.
- 2 They think that it is cleaner and safer to bury garbage in the ground.
- 3 They believe that it is cheaper and better for the environment.
- 4 They hope that it will provide more jobs for people in the city.

(38)

Which of the following statements is true?

- 1 A majority of people in Cairo were poor during the 1950s.
- 2 More young Zabbaleen now have the chance to attend university.
- 3 Cities with over 15 million people have found a cheaper way to collect garbage.
- 4 European companies can collect garbage at half of the cost of Egyptian companies.